

SEM.	CODE	COURSE TITLE	Prerequisite	Course Category	Theory Credit	Lab Credit	Sem.-wise Credit
1st (1-1)	SE 111	Computer Fundamentals		CORE	3		16
	SE 112	Computer Fundamentals Lab		LAB		1	
	SE 113	Introduction to Software Engineering		CORE	3		
	ENG 114/ ENG 101	English Reading, Writing Skills & Public Speaking/ English I		GED	3		
	GE 115/ AOL 101	Art of Living		GED	3		
	MAT 124/ MAT 101	Math-I: Calculus & Geometry/ Mathematics I		GED	3		
2nd (1-2)	SE 121	Structured Programming	SE111, SE112	CORE	3		19
	SE 122	Structured Programming Lab	SE111, SE112	LAB		1	
	SE 123	Discrete Mathematics		GED	3		
	SE 212	Software Requirement Specifications & Analysis	SE113	CORE	3		
	SE 213	Digital Electronics & Logic Design	SE111, SE112	GED	3		
	PHY 125/ PHY 101	Physics: Mechanics, Electromagnetism & Waves/ Physics I		GED	3		
	MAT 134/ MAT 102	Math-II: Linear Algebra & Fourier Analysis/ Mathematics II	MAT124/ MAT101	GED	3		
3rd (2-1)	SE 131	Data Structure	SE121, SE122, SE 123	PGC	3		19
	SE 132	Data Structure Lab	SE121, SE122, SE 123	LAB		1	
	SE 133	Software Development Capstone Project	SE121, SE122	LAB-P		3	
	SE 211	Object Oriented Concepts	SE121, SE122	CORE	3		
	SE 222	Computer Architecture	SE213	PGC	3		
	STA 135/ STA 101	Probability & Statistics in Software Engineering/ Statistics I		GED	3		
	GE 314/ BNS 101	Bangladesh Studies		GED	3		
4th (2-2)	SE 214	Algorithms Design & Analysis	SE131, SE132	PGC	3		21
	SE 215	Algorithms Design & Analysis Lab	SE131, SE132	LAB		1	
	SE221	Object Oriented Design	SE211	CORE	3		
	SE 223	Database System		CORE	3		
	SE 224	Database System Lab		LAB		1	
	SE 232	Operating System & System Programming	SE222	PGC	3		
	SE 233	Operating System & System Programming Lab	SE222	LAB		1	
	GE 235	Principles of Accounting, Business & Economics		GED	3		
	SE 532/ SE XXX	Introduction to Robotics (G-1)/ Guided Elective- I	SE213	PGC	3		
5th (3-1)	SE 225	Data Communication & Computer Networking		PGC	3		20
	SE 226	Data Communication & Computer Networking Lab		LAB		1	
	SE 231	System Analysis & Design Capstone Project	SE133, SE212, SE221	LAB-P		3	
	SE 234	Theory of Computing		PGC	3		
	SE 311	Design Pattern	SE221	CORE	3		
	SE 312	Software Quality Assurance & Testing	SE212, SE221	CORE	3		
	SE 313	Software Quality Assurance & Testing Lab	SE212, SE221	LAB		1	
	GE 324	Business Analysis & Communication	ENG114/ ENG101, SE212	GED	3		
6th (3-2)	SE 321	Software Engineering Web Application	SE121, SE122	CORE	3		20
	SE 322	Software Engineering Web Application Lab	SE121, SE122	LAB		1	
	SE 323	Software Architecture & Design	SE212, SE221, SE223, SE311	CORE	3		
	SE 332	Information System Security		CORE	3		
	SE 411	Software Project Management & Documentation	SE312	CORE	3		
	SE 333	Artificial Intelligence	SE214, SE234	CORE	3		
	SE 334	Artificial Intelligence Lab	SE214, SE234	LAB		1	
	SE 544/ SE XXX	Introduction to Machine Learning (G-2)/ Guided Elective- II	MAT 101,MAT 102, STA 101, SE 121	PGC	3		
7th (4-1)	SE 331	Software Engineering Design Capstone Project	SE231, SE312, SE321, SE323	LAB-P		3	21
	GE 557/ GED 302/ EMP101	Employability 360 Non-Major Only		GED	3		
	SE 599/ SE XXX	Research Methodology & Scientific Writing (Open Elective-3) Non-Major Only		PGC	3		
	SE 442/ SE XXX	Management Information System (G-3)/ Guided Elective- III		PGC	3		
	SE 535/ SE XXX	Data Warehouse & Data Mining (Non-Major Only)		PGC	3		
	RE 411	Embedded Systems Design & Development (RE Major Only)	RE 331, RE 332	MAJOR	2		
	RE 412	Embedded Systems Design & Development Lab (RE Major Only)	RE 331, RE 332	MAJOR-L		1	
	RE 421	Robotic Process Automation Design & Development (RE Major Only)	RE 331, RE 332, RE 411,RE 412,SE 442, SE 331	MAJOR	2		
	RE 422	Robotic Process Automation Design & Development Lab (RE Major Only)	RE 331, RE 332, RE 411,RE 412,SE 442, SE 331	MAJOR-L		1	
	CS 211	Cyber Security Fundamentals (CS Major Only)	SE 332	MAJOR	3		
	CS 418	Network & Communication Security (CS Major Only)	CS 211	MAJOR	3		
	DS 331	Introduction To Data Science and Data Management & Analysis (DS Major Only)	STA 101, SE 121	MAJOR	2		
	DS 332	Introduction To Data Science and Data Management & Analysis Lab (DS Major Only)	STA 101, SE 121	MAJOR-L		1	
	DS 411	Statistical Data Analysis (DS Major Only)	DS 331, DS 332	MAJOR	2		
	DS 412	Statistical Data Analysis Lab (DS Major Only)	DS 331, DS 332	MAJOR-L		1	
	SE 431	Final Year Project/ Thesis/ Internship		PRO-THE		6	
8th (4-2)	SE XXX	Guided Elective-IV Non Major Only		PGC	3		09
	SE 441	Software Engineering Professional Ethics		PGC	3		
	SE341/ SE XXX	Numerical Analysis (G -5) Non -Major Only/ Guided Elective- V		PGC	3		
	SE 447/ SE XXX	Human Computer Interaction (Open Elective-2) Non-Major Only		PGC	3		
	RE 331	Embedded Programming (RE Major Only)	SE 213, SE 121, SE 532	MAJOR	2		
	RE 332	Embedded Programming Lab (RE Major Only)	SE 213, SE 121, SE 532	MAJOR-L		1	
	RE 423	Advanced Robotics (RE Major Only)	RE 331, RE 332, RE 411,RE 412	MAJOR	2		
	RE 424	Advanced Robotics Lab (RE Major Only)	RE 331, RE 332, RE 411,RE 412	MAJOR-L		1	
	CS 422	Digital Forensic (CS Major Only)	CS 418	MAJOR	3		
	CS 334	Ethical Hacking & Countermeasures (CS Major Only)	CS 418	MAJOR	1		
	CS 335	Ethical Hacking & Countermeasures Lab (CS Major)	CS 418	MAJOR-L		2	
	DS 421	Machine Learning Driven Data Analysis I(DS Major Only)	DS 411,DS 412, SE 544	MAJOR	2		
	DS 422	Machine Learning Driven Data Analysis Lab I (DS Major Only)	DS 411,DS 412, SE 544	MAJOR-L		1	
	DS 423	Machine Learning Driven Data Analysis II and Communicating Data Insights (DS Major Only)	DS 411,DS 412, SE 544	MAJOR	2		
	DS 424	Machine Learning Driven Data Analysis II and Communicating Data Insights Lab (DS Major Only)	DS 411,DS 412, SE 544	MAJOR-L		1	
	RE 431	Robotics & Embedded Systems Major Capstone Project (RE Major Only)	ALL RE Major courses	PRO-THE		6	
	CS 439	Cyber Security Major Capstone Project (CS Major Only)	ALL CS Major Courses	PRO-THE		6	
	DS 431	Data Science Major Capstone Project (DS Major Only)	ALL DS Major Courses	PRO-THE		6	
		TOTAL CREDIT			116-120	25-29	145
		TOTAL COURSE					

* Course Categories:

1. GED: General Education Courses
2. PGC: Preparatory General Core
3. CORE: Software Engineering Core
4. LAB: Laboratory Courses
5. LAB-P: Lab Based Project Courses
6. MAJOR: Major Courses (All Majors)
7. MAJOR-L: Major Laboratory Courses (All Majors)
8. PRO-THE: Final Year Project/ Thesis